

Article

Hydrodynamic Analysis of Different Formation Configurations of Catamaran in Regular Head Waves

Zhifan Zhang ¹, Bo Jiang ¹, Longkan Wang ², Shengren Wei ^{1,3}, Tao Li ⁴, Guiyong Zhang ^{1,*}  and Zhi Zong ⁵¹ State Key Laboratory of Structural Analysis, Optimization and CAE Software for Industrial Equipment, School of Naval Architecture Engineering, Dalian University of Technology, Dalian 116024, China² China Ship Research and Development Academy, Beijing 100192, China³ Dalian Shipbuilding Industry Co., Ltd., Dalian 116011, China⁴ School of Mechanical Engineering, Key Laboratory of Advanced Intelligent Protective Equipment Technology, Ministry of Education, Hebei University of Technology, Tianjin 300401, China⁵ Fuyao University of Science and Technology, Fuzhou 350100, China

* Correspondence: gygzhang@dlut.edu.cn

Abstract: When undertaking long-distance missions at sea, vessels aim to achieve an extended operational range through drag reduction and energy efficiency, while enhanced wave resilience also provides substantial benefits. In this work, the Delft-372 catamaran is utilized to investigate the feasibility of drag reduction and roll mitigation for catamaran formation sailing in waves, analyzing the effects of three different formation configurations and varying spacings. The overset grid method was employed to simulate vessel motions, while the Volume of Fluid (VOF) method captured the free surface. First, the numerical results of the catamaran's resistance, pitch, and heave motion amplitudes under different wave conditions were compared with experimental data to verify the accuracy of the CFD numerical method, and a grid convergence analysis was performed. Next, numerical models of the Delft-372 catamaran were constructed in parallel, tandem, and lateral formations under wave conditions. The results of the single-ship simulation were employed as a benchmark to analyze the impact of different formation configurations and varying lateral and longitudinal spacings on the resistance, pitch, and heave motions of the catamarans. The study also examined the effects of wave interference between vessels and the combined influence of external waves on individual and overall hydrodynamic performance. Results indicated that the tandem formation outperformed the parallel and lateral formations, with optimal performance observed at the longitudinal distance of $1 L_{pp}$. Generally, during navigation, the follower catamaran should ideally be positioned in the trough of the stern wave of the leader catamaran.

Keywords: formation configuration; catamaran; hydrodynamic optimization; CFD

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1. Introduction

In recent years, unmanned catamarans have been widely used in civil and military applications due to their excellent stability and seaworthiness. As research on single-ship control problems matures [1] and intelligent ships and unmanned vessels develop, ship formation sailing has emerged as an inevitable trend in future ship development. Ship formation sailing can significantly reduce the resistance of ship navigation, which is significant in reducing energy consumption and enhancing shipping efficiency. It has broad application prospects in areas such as ship cluster operations and maritime search and rescue.

Nowadays, various risks, such as global warming, are caused by the increased greenhouse gas effect on the atmosphere [2]. The shipping industry is considered one of the largest sources of greenhouse gas emissions, including carbon dioxide. Reducing resistance to save energy and cut emissions can effectively mitigate this phenomenon. However, traditional methods of reducing resistance, such as hull improvements, have been extensively

optimized. Are there other methods available to further reduce resistance? By studying behaviors, such as the flocking of wild geese and ducks (Figure 1), researchers have gradually discovered another form of drag reduction: formation sailing. Various formation concepts have been proposed, leading to in-depth research on the effects of hydrodynamic forces on ship motion.

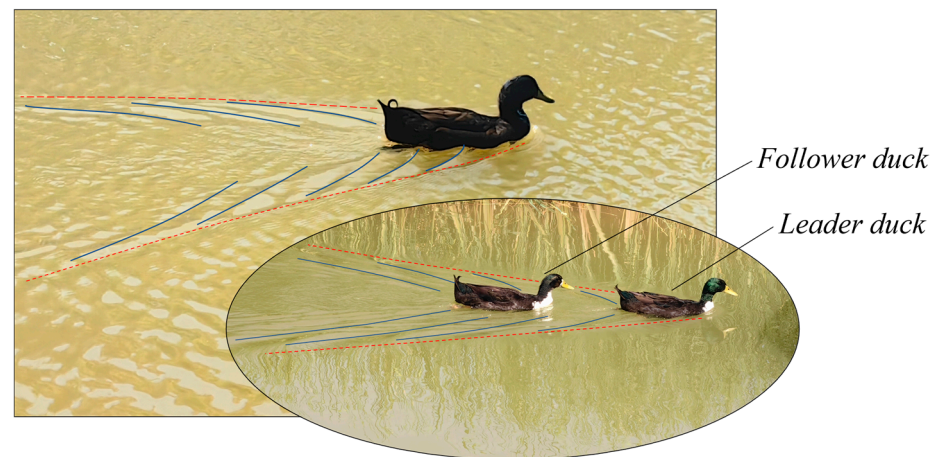


Figure 1. The flocking of ducks.

For surface ships, some pilot attempts have been made to optimize navigation strategies. The NOVIMAR project [3] developed an innovative concept of waterborne platooning, called the Vessel Train [4]. He et al. [5] conducted research to optimize drag reduction in three different formation configurations: tandem, parallel, and triangle, and provided the optimal arrangement and spacing. Jiang et al. [6] analyzed the navigation process of two-ship formations in the brash ice channel, examining the hydrodynamic forces on the ship, ship–ice interaction, and ice resistance effect. Results showed that the ice resistance of the rear ship decreased as the distance decreased. Dong et al. [7] researched energy-saving optimization for Unmanned Surface Vessel (USV) formations, mainly focusing on analyzing resistance changes to determine the optimal formation spacing for minimal energy consumption. Tian et al. [8] proposed a method to optimize the layout of Autonomous Underwater Vehicle (AUV) formations by analyzing the hydrodynamic resistance of two AUVs at different positions and deriving the optimal layout parameters. Zhou et al. [9] summarized the hydrodynamic analysis of ship–ship interactions over the last decade. Ma et al. [10] used a four-point model to consider the interference between the bow and stern waves of a monohull, as well as the interference between two ships, and investigated the transverse wave characteristics and the wave resistance at medium and high Froude numbers. The relationship between the spacing between the two ships and the Fr , which reduces the drag of the rear ship, was provided.

Furthermore, numerical methods save manpower and costs compared to theoretical and experimental approaches [11–13]. Viscous methods, such as the Reynolds-averaged Navier–Stokes (RANS) method, are used to analyze the effects of ship-to-ship interactions in the process of head-on, collision, and overtaking [14,15], the interference of passing ships with moored ships [16,17], two ships moving in parallel during replenishment or discharge operations [18], and tugs escorting or assisting large ships in berthing, platforming, or steering movements [19].

Research on catamarans has been thriving both domestically and internationally. Farkas et al. [20] conducted a numerical study to analyze the hydrodynamic interference between the S60 catamaran and a monohull, exploring the catamaran's interference resistance characteristics, resistance components, and the relationship between form factor and spacing. Martić et al. [21] employed the STAR-CCM+ software, integrating the RANS $k - \omega$ SST turbulence model, to numerically simulate the impact of shallow water conditions on the total resistance of the SolarCat solar-powered catamaran. Wang et al. [22] experimen-

tally and numerically studied the planing catamaran hull or three displacements in calm water. Ali et al. [23] conducted research based on the Delft-372 catamaran and concluded that wave disturbance factors are important for the catamaran. Benjamin et al. [24] studied the seakeeping of a high-speed catamaran in head waves, providing a valuable database for the hydrodynamic research and CFD verification of the high-speed catamaran.

In summary, studying the hydrodynamic effects of vessel formation sailing can optimize formation configuration and navigation strategies, minimizing resistance, reducing fuel costs, and extending range. Numerous scholars worldwide have long been deepening their research on vessel formation sailing, continually optimizing and innovating related control methods and formation layouts. However, most current research on the hydrodynamic effects of formation sailing focuses on monohulls and is primarily conducted in calm water conditions. There is a lack of research on the hydrodynamic interactions between catamarans and even multihulls, especially under wave conditions. Catamarans exhibit superior wave resistance compared to monohulls, but their wake effects are more complex. To investigate the combined effects of wave interference between catamarans and external waves, the novelties of this study are as follow:

1. This study examined the hydrodynamic effects of two catamarans sailing in formation in waves, including resistance, pitch, and heave motions. The study analyzed the combined influence of wave interference between the vessels and external waves on individual and overall hydrodynamic performance.
2. Three different formation configurations formed by two catamarans: parallel, tandem, and lateral formation, were investigated. The wake effects and flow field characteristics of catamarans under different formation configurations were analyzed to determine the formation with the optimal drag reduction and roll mitigation effects.
3. The study also examined the impact of different lateral and longitudinal distances on the hydrodynamics of the formation, determining the optimal transverse and longitudinal spacings for each formation configuration.

When undertaking long-distance missions at sea, vessels aim to achieve extended operational range through drag reduction and energy efficiency, while enhanced wave resilience also provides substantial benefits. In this work, the Delft-372 catamaran is utilized to investigate the feasibility of drag reduction and roll mitigation for catamaran formation sailing in waves, analyzing the effects of three different formation configurations and varying spacings. The overset grid method is employed to simulate vessel motions, while the Volume of Fluid (VOF) method captures the free surface. The specific work is as follows. First, the numerical results of the catamaran's resistance, pitch, and heave motion amplitudes under different wave conditions are compared with experimental data to verify the accuracy of the CFD numerical method, and a grid convergence analysis is performed. Next, numerical models of the Delft-372 catamaran are constructed in parallel, tandem, and lateral formations under wave conditions. The results of the single-ship simulation are employed as a benchmark to analyze the impact of different formation configurations and varying lateral and longitudinal spacings on the resistance, pitch, and heave motions of the catamarans. The study also examines the effects of wave interference between vessels and the combined influence of external waves on individual and overall hydrodynamic performance. The optimal formation configurations and spacings are identified, and how to utilize beneficial wave interference is discussed. The Delft-372 catamaran used in this study has extensive experimental data available for reference, which can serve as a basis for future research on wave interference between other catamarans and even multihulls.

2. Numerical Theory

2.1. Control Equations

In this study, the commercial CFD software STAR-CCM+ (17.04) was used. The unsteady RANS model of STAR-CCM+ and the VOF method were employed to capture the free surface of the liquid and handle the two-phase or multiphase flow. The fluid's continuity momentum equations [25] are presented in Equations (1) and (2).

The continuity equation is given by [25]:

$$\frac{\partial \rho}{\partial t} + \frac{\partial(\rho U_i)}{\partial x_i} = 0 \quad (1)$$

where, ρ is the fluid density, t is the time, x_i ($i = 1, 2, 3$) represents the spatial coordinates in 3 directions, and U_i is the component of the velocity vector U in the i direction.

The momentum equation is expressed as [25]:

$$\frac{\partial(\rho U_i)}{\partial t} + \frac{\partial(\rho U_i U_j)}{\partial x_j} = -\frac{\partial P}{\partial x_i} + \frac{\partial \tau'}{\partial x_j} + \frac{\partial(\rho \overline{U_i U_j})}{\partial x_j} + F_i \quad (2)$$

where, U_j is the component of the velocity vector U in the j direction, x_j ($j = 1, 2, 3$) represents the spatial coordinates in the 3 directions, P is the pressure, and τ' is the mean component of the viscous stress tensor.

In determining the position of the free level, all fluids need to follow the law of conservation of mass, written as [25]:

$$\frac{\partial}{\partial t} r_\omega \rho_\omega \vec{V} + \nabla \cdot r_\omega \rho_\omega \vec{V} = 0 \quad (3)$$

$$\frac{\partial}{\partial t} r_a \rho_a \vec{V} + \nabla \cdot r_a \rho_a \vec{V} = 0 \quad (4)$$

$$r_\omega + r_a = 1 \quad (5)$$

where, ρ_ω and ρ_a are the densities of water and air, respectively, while r_ω and r_a are the volume fractions of water and air, respectively.

2.2. Turbulence Modeling

The turbulence modeling [26] provides cyclonic corrections for rotating flow and flow separation with good convergence and accuracy.

The turbulent kinetic energy equation is given by [24]:

$$\frac{\partial(\rho k)}{\partial t} + \frac{\partial(\rho k U_j)}{\partial x_j} = \frac{\partial[(\mu + \frac{\mu_t}{\sigma_k}) \partial k / \partial x_j]}{\partial x_j} + G_k + G_b - \rho \varepsilon - Y_M \quad (6)$$

where, k is the turbulent kinetic energy, ρ is the fluid density, t is the time, U_j is the component of the velocity vector U in the j direction, x_j ($j = 1, 2, 3$) represents the spatial coordinates in the 3 directions, μ is the dynamic viscosity coefficient, ε is the turbulent kinetic energy dissipation rate, σ_k is a constant, G_k is the turbulent kinetic energy generated by laminar flow, G_b is the turbulent kinetic energy generated by buoyancy, and Y_M is the fluctuation generated by transition diffusion in compressible turbulence.

The turbulent dissipation rate equation is expressed as [24]:

$$\frac{\partial(\rho \varepsilon)}{\partial t} + \frac{\partial(\rho \varepsilon U_j)}{\partial x_j} = \frac{\partial[(\mu + \frac{\mu_t}{\sigma_\varepsilon}) \partial \varepsilon / \partial x_i]}{\partial x_i} + \rho C_1 \bar{S} \varepsilon - \frac{C_2 \rho \varepsilon^2}{k + \sqrt{\nu \varepsilon}} + \frac{C_{\varepsilon 1} C_{\varepsilon 3} C_b \varepsilon}{k} \quad (7)$$

where, $\bar{S}$ is the average strain rate, and σ_s , C_1 , C_2 , $C_{\varepsilon 1}$, and $C_{\varepsilon 3}$ are all constants.

The realizable $k - \varepsilon$ two-equation turbulence model incorporates swirl modifications, rendering it particularly effective for simulating rotating flows and flow separation with superior convergence and accuracy. This model is exceptionally proficient in handling complex geometries and abrupt directional shifts in flow dynamics. Implementing the realizable $k - \varepsilon$ model enables more precise predictions of catamaran performance under varying inter-vessel spacings in formation.

2.3. Numerical Modeling

The Finite Volume Method (FVM) is a widely adopted numerical approach extensively utilized in fluid dynamics and thermodynamics. Its core principle revolves around discretizing the computational domain by applying conservation laws in their integral form. Within the FVM framework, the computational domain is systematically partitioned into a series of non-overlapping control volumes (or cells), and the conservation laws for mass, momentum, and energy are applied to each individual control volume. Conservation equations are established on each control volume and subsequently discretized into a system of algebraic equations, thereby yielding a solvable system. This process typically entails selecting an appropriate discretization scheme (e.g., central differencing or upwind differencing) to approximate derivative terms and employing iterative methods (e.g., SIMPLE or PISO algorithms) to solve the nonlinear equations. Owing to its inherent conservation and stability properties, FVM is highly effective in addressing complex fluid dynamics problems.

The Volume of Fluid (VOF) method is a numerical technique designed to track and simulate the motion of interfaces between two immiscible fluids, particularly air–water interfaces or free surfaces. The core concept of the VOF method lies in representing the interface within each computational cell as a mixture of two fluids, using volume fractions to indicate the relative amounts of each fluid present. During the simulation, the volume fraction is transported through the computational domain along with the fluid flow, enabling the accurate tracking of the interface's position and shape over time. The computational domain is divided into multiple cells (or grids), with a volume fraction (α) used to represent the presence of both fluids within each cell. The volume fraction, α , represents the proportion of a specific fluid within a cell, ranging from 0 to 1, where 0 indicates that the cell is entirely occupied by the other fluid, and 1 indicates full occupancy by the designated fluid. When $0 < \alpha < 1$, it signifies that the cell contains a mixture of the two fluids, indicating that the interface traverses the cell. The VOF method can accurately track and simulate the morphology of free surfaces, maintaining high precision even under complex flow conditions. Due to its reliance on the concept of volume fractions, the VOF method can handle complex interface deformations, such as interface break-up and merging, without concerns about instability issues related to surface reconstruction. For flow problems involving free surfaces, such as ship navigation and wave interactions, the VOF method offers a powerful tool to simulate these complex physical phenomena without the need for explicit surface reconstruction. Compared to other methods, such as the Level Set and Front Tracking methods, the VOF method provides superior stability in handling free surface flows, particularly suited for long-duration simulations and complex interface deformations.

Overset grids, also referred to as nested grids, are a commonly employed technique in fluid dynamics simulations involving multiple objects or moving bodies. The fundamental principle involves partitioning the computational domain into multiple, independent grid regions that overlap in physical space, allowing each grid to be independently generated and adjusted according to object motion. Key features of this technique include flexible grid partitioning, efficient interpolation of physical quantities, adaptability to complex motions, and automated handling of grid interactions. When utilizing overset grids, the computational domain is typically divided into two regions: a background region and an overlapping region. The background region covers the entire computational domain, while the overlapping region is slightly larger than the moving object, fully encapsulating it. During the simulation, these two regions are further subdivided into three types of cells: active cells, donor (or receiver) cells, and inactive cells. Active cells handle the discretization and solution of governing equations, donor (or receiver) cells, located near the overlapping boundaries, manage the interpolation and transfer of physical quantities between regions, and inactive cells are temporarily removed from the computation. The primary advantages of this technique include its simple grid geometry, which accurately reflects model motion, enhanced computational efficiency and reduced modeling workload, simplified boundary

condition settings, and high grid quality, which remains unaffected by object position and orientation. These attributes make overset grids highly effective in simulating complex motion problems.

3. Numerical Model

3.1. Geometric Modeling

The Delft-372 catamaran model, developed by Delft University of Technology (DUT) in the Netherlands [27], was utilized in this study, as depicted in Figure 2. The relevant dimensions of the Delft-372 catamaran model are listed in Table 1.

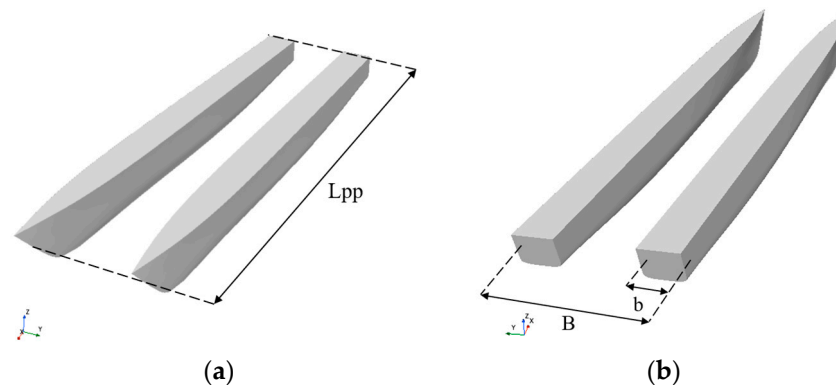


Figure 2. The 3D model of the Delft-372 catamaran: (a) bow view and (b) stern view.

Table 1. Dimensions of the Delft-372 catamaran.

Parameters	Symbol (Units)	Value
Length between perpendiculars	L_{PP} (m)	3.000
Beam demihull	b (m)	0.240
Beam overall	B (m)	0.940
Draught	T (m)	0.150
Displacement	Δ (kg)	87.070
Distance between center of hulls	H (m)	0.700
Longitudinal center of gravity	X_g (m)	1.410
Vertical center of gravity	Z_g (m)	0.340
Longitudinal moment of inertia	K_{yy} (m)	0.782

3.2. Initial Condition of Numerical Model

The initial consideration is the application of verification calculations using a single catamaran. The size of the computational domain in this section was determined based on a single catamaran, with appropriate adjustments made when applying to scenarios involving two catamarans, while keeping other settings consistent. Overset mesh technology was employed in the numerical simulation of a ship sailing in waves. Before meshing, the complex watershed must be divided into the background domain and the overset domain. The overset domain should be capable of containing the entire hull and cooperating with its movements. When the overset domain moves, it forms an interface with the background domain, facilitating information exchange between the grids. In this study, a rectangular cuboid computing domain was utilized, as illustrated in Figure 3a,b. The front end of the computational domain was $1.5 L_{PP}$ from the bow, while the back end was $4 L_{PP}$ from the stern. The width of the computational domain was $3 L_{PP}$, and the depth of the computational domain was $3.5 L_{PP}$. The height of the computational domain's bottom was set to twice the ship's length from the keel, ensuring that the bottom of the domain did not interfere with changes in the flow beneath the ship or the formation and development of wave patterns, thereby minimizing the impact of boundary conditions on the simulation results. However, as the distance increased, computational costs also rose, necessitating a balance between accuracy and computational efficiency. Generally, a distance of twice

the ship's length is considered reasonable, with further increases having minimal impact on the results. Of the six boundaries, only the rear boundary behind the stern was set as a pressure outlet, while the remaining five boundaries (top, front, bottom, left, and right surfaces) were configured as velocity inlets. The boundary conditions for all cases in this study remained consistent.

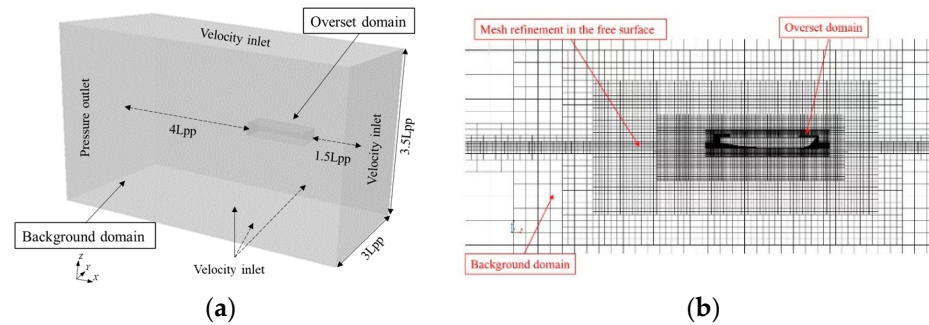


Figure 3. Overset domain and background domain. (a) Regional division and (b) mesh around the slice and mesh differences between regions.

To ensure accurate information exchange between the overlapping and background regions, the mesh size of these regions must be consistent. The region and the surface grid of the ship are shown in Figure 4a–c.

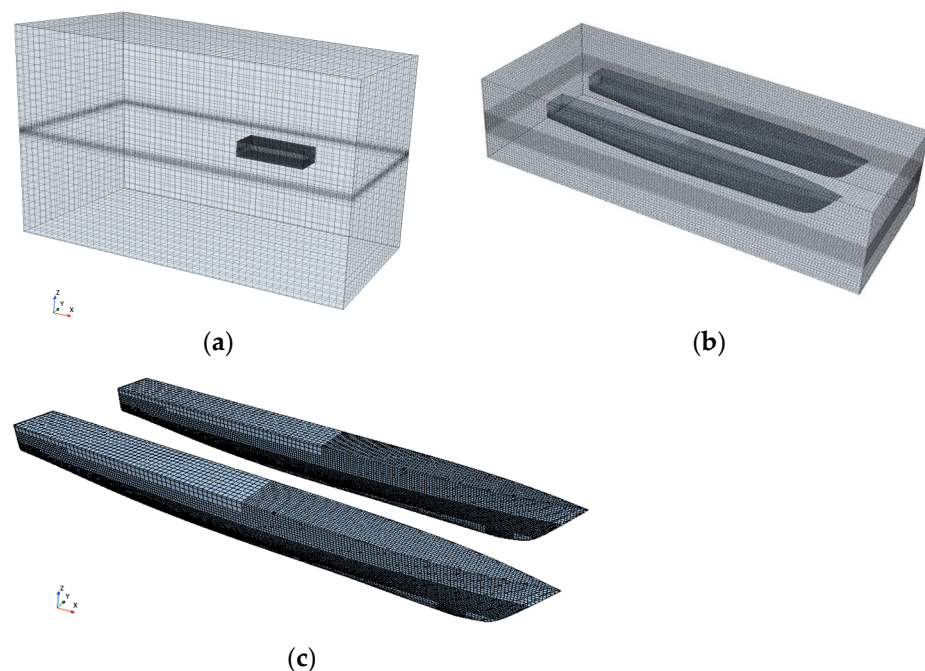


Figure 4. Domain and hull grid. (a) Computational domain mesh, (b) overset domain mesh, and (c) hull surface mesh.

3.3. Verification of Grid Independence

Ensuring the reliability of numerical results is crucial, particularly when addressing the uncertainties inherent in the mathematical models used to solve CFD problems. The impact of these uncertainties on the numerical model should be assessed through validation studies, with one of the primary sources of uncertainty being the grid structure. Grid-related uncertainties should be evaluated using the Grid Convergence Index (GCI) method [28], which requires constructing at least three sets of uniformly refined grids (fine, medium, and coarse), corresponding to grid numbers N_1 , N_2 , and N_3 . This study focused primarily

on investigating the resistance and seakeeping performance of the Delft-372 catamaran in waves. Consequently, the total resistance in calm water was selected as a key parameter, while resistance in waves was chosen for seakeeping analysis.

The wavelength, λ , was set to 6 m, and $\lambda/L_{PP} = 2$. In this study, a force dissipation method was selected, with the dissipation zone length defined as 1.5 m. The Froude number was set to 0.5, and the time step was set to 0.008 s. The iteration time step and the number of grid-near-wall-prism layers were set to 10. The ship model was permitted to undergo pitch and heave motions.

The refinement factors of the grids r_{21} and r_{23} , and the order of discretization p , are defined as:

$$r_{21} = \sqrt[3]{\frac{N_1}{N_2}}, r_{23} = \sqrt[3]{\frac{N_2}{N_3}} \quad (8)$$

$$p = \frac{1}{\ln(r_{21})} |\ln\left(\frac{\varepsilon_{32}}{\varepsilon_{21}}\right)| + q(p) \quad (9)$$

$$q(p) = \ln\left(\frac{r_{21}^p - s}{r_{32}^p - s}\right) \quad (10)$$

$$s = 1 \cdot \text{sgn}\left(\frac{\varepsilon_{32}}{\varepsilon_{21}}\right) \quad (11)$$

where $\varepsilon_{21} = \phi_2 - \phi_1$, $\varepsilon_{32} = \phi_3 - \phi_2$, and ϕ_i are the simulation results for different numbers of grids. When $s = 1$, it indicates that the results showed consistent convergence, and $s = -1$ indicates that the results showed oscillatory convergence. Extrapolated values, ϕ_{ext}^{21} , are defined as:

$$\phi_{ext}^{21} = \frac{r_{21}^p \phi_1 - \phi_2}{r_{21}^p - 1} \quad (12)$$

Approximate errors, e_a^{21} , and extrapolated relative errors, e_{ext}^{21} , are defined as:

$$e_a^{21} = \left| \frac{\phi_2 - \phi_1}{\phi_1} \right| \quad (13)$$

$$e_{ext}^{21} = \left| \frac{\phi_{ext}^{12} - \phi_1}{\phi_{ext}^{12}} \right| \quad (14)$$

The convergence index, GCI_{fine}^{21} , is obtained by the following expression:

$$GCI_{fine}^{21} = \frac{1.25e_a^{21}}{r_{21}^p - 1} \quad (15)$$

Based on the discretization error components provided in Table 2 for the grid convergence study, the analysis of the Delft-372 in calm water and regular waves was found to be monotonically convergent. The numerical uncertainties in the resistance of the catamaran in calm water and regular waves were 0.382% and 1.202%, respectively, both under 2%.

Generally, smaller grid sizes result in a higher calculation accuracy but increase the computational time. To achieve an optimal balance between computational resources and accuracy, an appropriate grid size must be selected. After calculations, all three grid sets exhibited monotonic convergence, meeting the convergence criteria. It is concluded that the resistance error using a grid base size of 0.1 m was within 5%, meeting the required calculation accuracy. Considering the available computational resources and desired accuracy, this grid size was used for the subsequent simulation study.

Table 2. Calculation of the grid-independence verification of the Delft-372 catamaran.

Components	Resistance (Calm Water)	Resistance (Regular Waves)
Fine (No. of cells)	3,508,235	3,635,980
Medium (No. of cells)	2,025,182	2,308,464
Coarse (No. of cells)	1,118,578	1,415,715
r_{21}	1.201	1.163
r_{23}	1.219	1.177
ϕ_1 (N)	53.042	56.328
ϕ_2 (N)	52.780	55.945
ϕ_3 (N)	52.005	55.123
ε_{21}	−0.262	−0.383
ε_{32}	−0.775	−0.722
s	1	1
q	−0.124	−0.100
p	5.250	3.541
e_a^{21} (%)	0.494	0.680
GCI_{fine}^{21} (%)	0.382	1.202

3.4. Comparison of Simulation Test Results

Compared to a single ship, the catamaran exhibited superior resistance performance at high speeds. In this section, the high-speed condition of $Fr = 0.7$ was selected for numerical verification of seakeeping and wave-added resistance. A series of regular wave conditions were selected to match the EFD results presented in [29], as shown in Table 3. These data were based on seakeeping trials conducted on the Delft-372 catamaran in regular waves, which aligns with the conditions of this study. Comparison with experimental data allows for the validation of the numerical methods and models employed in this study. This calculation considered only the motion of heave and pitch. In this study, a force dissipation method was selected, with the dissipation zone length was defined as 1.5 m.

Table 3. Regular wave conditions of the Delft-372 catamaran.

Case	Froude Number, Fr	Wave/Ship Length, λ/L_{PP}	Wave Height, H (m)	Encounter Frequency, W_e (rad/s)	Encounter Period, T_e
1	0.7	1.00	0.06667	6.2431	0.50321
2	0.7	1.50	0.10000	4.5016	0.69788
3	0.7	1.75	0.11667	3.9856	0.78823
4	0.7	2.00	0.13333	3.5909	0.87487
5	0.7	2.25	0.15000	3.2783	0.95829

Based on the model test study by R. Broglia (2011) [29] at the Italian INSEAN tank, numerical simulations were conducted for the aforementioned conditions in this study. The comparison of the numerical non-dimensional additional resistance, pitch, and heave values caused by regular waves with the EFD results is shown in Figures 5 and 6. The motion responses of the catamaran in waves exhibited dynamic time dependence and simple harmonic motion, which were compatible with each other. Consequently, the discrepancy between EFD and CFD was within a tolerable range for seakeeping simulations.

According to the established definition, the wave-added resistance of the catamaran is the time-averaged total resistance of the catamaran over a complete period, minus its resistance in still water. The wave-added resistance coefficient, C_{aw} , of the catamaran is defined as [30]:

$$C_{aw} = \frac{R_{aw}}{\rho g \zeta_A^2 B^2 / L} \quad (16)$$

where ρ is the fluid density, g is the gravitational acceleration, ζ_A is the regular wave amplitude, B is the overall beam, and L is the length between perpendiculars.

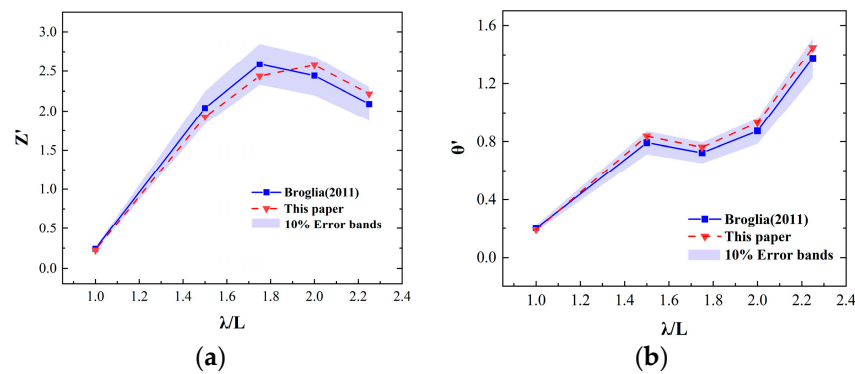


Figure 5. Heave and pitch motion responses of the Delft-372 catamaran with $Fr = 0.7$ in head waves. (a) Heave RAO and (b) pitch RAO [29].

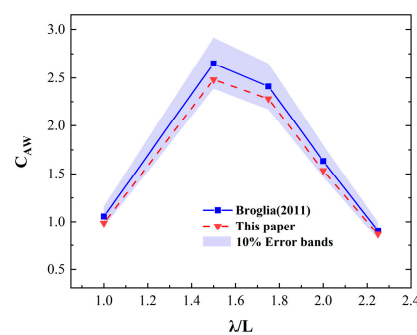


Figure 6. Added resistance coefficient of the Delft-372 catamaran with $Fr = 0.7$ in head waves [29].

The pitch RAO is expressed as [31]:

$$\theta' = \frac{\theta}{(360\zeta_A/\lambda)} \quad (17)$$

and the heave RAO can be expressed as:

$$Z' = \frac{Z}{\zeta_A} \quad (18)$$

where θ is the pitch amplitude, λ is the wavelength, and Z is the heave amplitude. The numerical results exhibited a pattern comparable to the experimental data presented in Figures 5 and 6. For regular wave conditions, wave-added resistance varied with changes in wavelength and wave height, peaking at $\lambda/L = 1.75$ with a maximum increase in resistance of 36%. Similarly, for heave motion, the most significant motion occurred at $\lambda/L = 1.75$. For pitch motion, the trend was a gradual increase as the wavelength-to-ship-length ratio increased. The numerical method employed in this paper was shown to be reliable, with a maximum error of 10%, as evidenced by the results in Figures 5 and 6.

4. Analysis of Drag Reduction in Different Formation Configurations

In this section, the Delft-372 catamaran was used as the primary vessel, and the effects of formation arrangements and sailing distance on resistance and seakeeping performance were analyzed. The feasibility of reducing the resistance and motion amplitude was demonstrated.

To explore the influence of different formation configurations, the arrangement shown in Figure 7 was adopted. First, the lateral distance, SP , was selected as $0.25B$ to $2B$ in parallel formation. Then, the longitudinal distance, ST , was set to $0.25 L_{PP}$ to $2 L_{PP}$ in tandem formation. Finally, the longitudinal distance and the lateral distance were set to 0.5 to $1.5 L_{PP}$ and $0.25B$ to $1B$ in lateral formation, respectively.

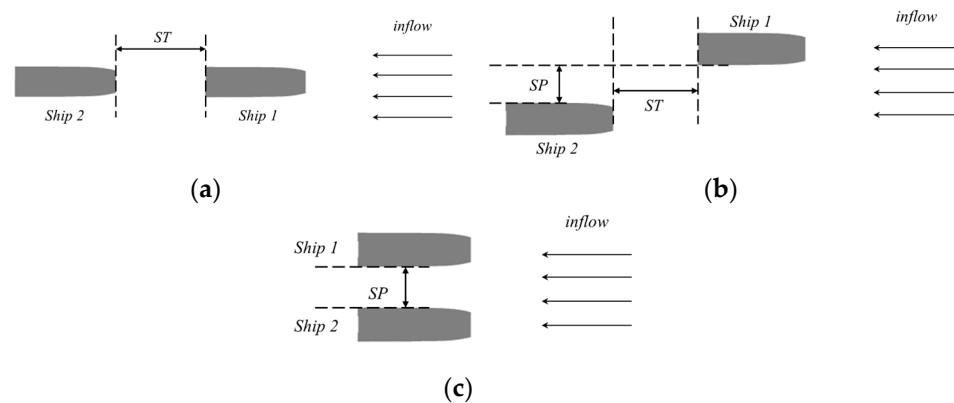


Figure 7. The geometric configuration of catamarans in different formations: (a) tandem formation, (b) lateral formation, and (c) parallel formation.

The boundary conditions and physical models were identical to those used in the previous section. The direct input method was used to simulate the fifth-order regular waves, with the wave conditions being combined with those used in [32]. In this section, the wavelength, λ , was set to 6 m, the wave height, ζ_A , to 0.15 m, and λ/ζ_A was 40. For the simulation test, the overlapping mesh and the DFBI technique were combined with the case of $Fr = 0.5$.

4.1. Parallel Formation

The total resistance coefficient and lateral force of the catamaran at different lateral distances are presented in Figure 8. The calculated total resistance coefficient for a single catamaran was 0.0076668. Due to the geometric symmetry, the resistance and motion responses of the two catamarans were almost identical. The maximum increase in the resistance coefficient was observed at the minimum lateral distance of $0.25 L_{PP}$, reaching 13.67%. When $SP \leq 1B$, the total resistance coefficient and lateral force of the catamaran exhibited a pronounced decline with the increasing lateral distance. When $SP \geq 1B$, both parameters exhibited a flattened trend with the increasing lateral distance, with the total resistance coefficient exhibiting an error within 5% of that of a single catamaran. Since the inflow in front of the catamaran remained relatively constant when the two catamarans were in parallel formation, and the inflow velocity remained the same, it can be concluded that the inflow in front of the catamaran did not significantly affect the resistance coefficient. Consequently, the resistance at each distance remained relatively constant.

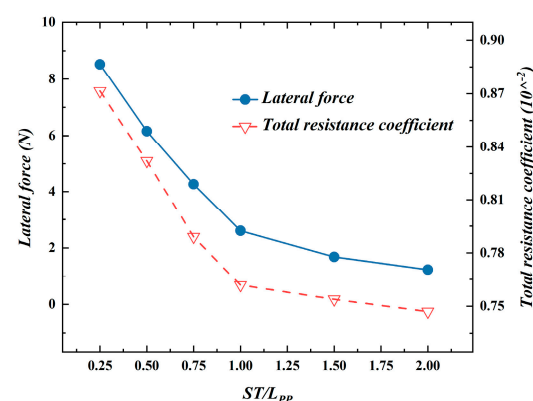


Figure 8. The resistance and lateral force of the catamaran under different lateral distances.

The heave and pitch amplitudes of the catamaran at different lateral distances are shown in Figure 9. This conclusion is further supported by the following evidence. It can be observed that the interaction forces between the two catamarans had minimal effect when the lateral distance exceeded $1B$.

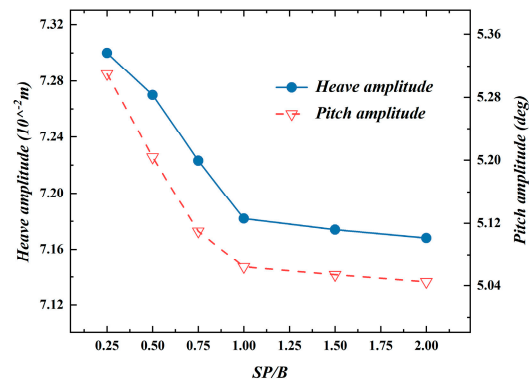


Figure 9. The heave and pitch amplitudes of the catamaran under different lateral distances.

The lateral distance significantly influenced the lateral force of the catamaran. The lateral force continuously increased as the lateral distance decreased. As the distance between the two catamarans decreased, the pressure gradient on their sides increased, the flow around the catamarans became asymmetric, and a pressure difference arose between the left and right sides. This pressure difference increased the suction between the two ships, thereby increasing the lateral force. Figure 10 illustrates the variations in the free surface and flow field around the hull at different lateral distances. It can be observed that when the lateral distance was small, the wakes of the two catamarans interacted, forming large areas of wave crests and troughs, with the converged wake crest located relatively close to the stern. The flow field was concentrated, with a limited range of diffusion. The intense changes in the flow field created significant suction effects between the two catamarans. The hydrostatic pressure distribution around the hull changed, and the wave loads resulting from ship interaction increased, leading to higher resistance and motion responses for the catamarans. As the lateral distance increased, the generated wave systems of the two catamarans began to separate, reducing the wave interference, and the converged wake crest moved further away from the stern. The wave pattern on the free surface gradually approached that of a single catamaran, with the wave height and wavelength stabilizing. Due to the reduced wave interference, the wave loads between the catamarans decreased accordingly. Thus, the distance between the two catamarans should be sufficient to ensure navigational safety and prevent ship suction, according to the safety distance theory of ship encounters. It is recommended that the distance between two ships be at least $1B$.

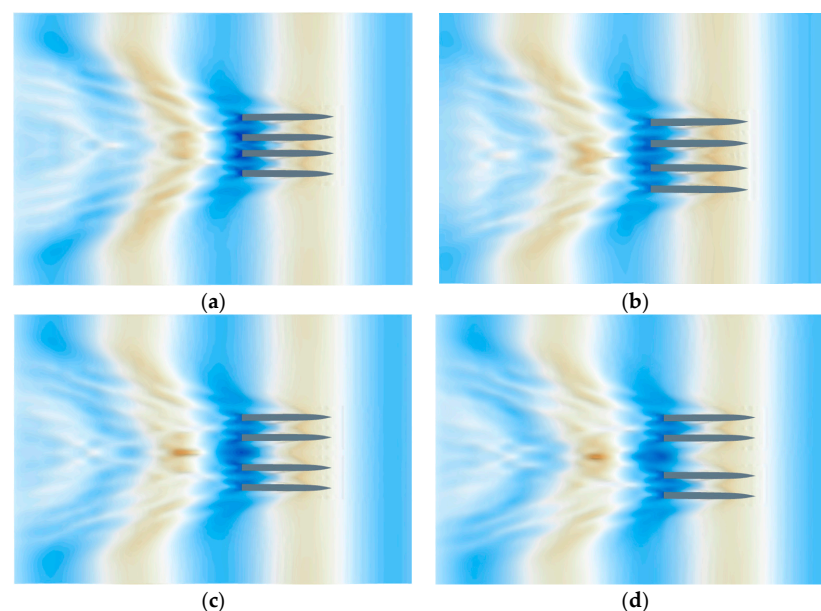


Figure 10. Cont.

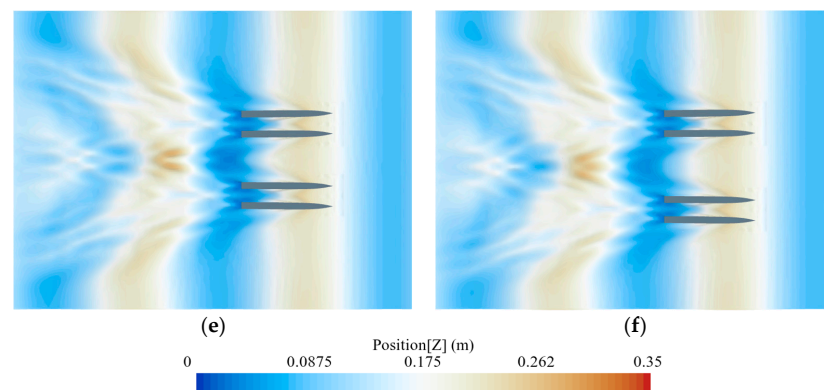


Figure 10. The free surface deformations of parallel formation at different lateral distances: (a) $SP/B = 0.25$, (b) $SP/B = 0.5$, (c) $SP/B = 0.75$, (d) $SP/B = 1$, (e) $SP/B = 1.5$, and (f) $SP/B = 2$.

4.2. Tandem Formation

In tandem formation, the interaction between the hulls had little effect on the lateral force and yaw moment, so the resistance, pitch, and heave motions were mainly analyzed. To study the influence of longitudinal distance on the hydrodynamic performance of ships, numerical simulations were conducted on two and three catamarans.

Figure 11 shows the comparison of total resistance coefficients for the leader and follower catamarans at different longitudinal distances with that of a single catamaran. The respective comparisons of pitch and heave amplitudes are shown in Figure 12a,b. The graphs show that the smaller variations in resistance and motion responses of the leader catamaran were similar to those of a single catamaran. For the follower catamaran, a reduction in both pitch and heave motion responses was observed with the increasing longitudinal distance, manifested as significantly reduced amplitudes. The resistance trend exhibited by the follower catamaran was shown to follow a “V-shape” pattern as a function of the longitudinal distance. The results indicate that the resistance, pitch, and heave motions of the leading catamaran were nearly identical to those of a single catamaran, suggesting that the sequential progression of the two catamarans had minimal impact on the leading catamaran’s navigation. When the longitudinal spacing was small, the resistance of the following catamaran was greater than that of a single catamaran. When the $ST = 0.25\text{--}0.5 L_{pp}$, the resistance of the following catamaran increased by approximately 44.66% and 32.18%, respectively. As the longitudinal distance increased, a pronounced drag reduction effect was observed on the follower catamaran, reaching a maximum of 36.61% at $ST = 1 L_{pp}$. Subsequently, as the longitudinal distance increased, the resistance of the following catamaran continued to rise, and when ST reached $1.75 L_{pp}$, the resistance surpassed that of a single catamaran. It is evident that the position of the following catamaran relative to the leading catamaran influenced the resistance. When the waves generated by the leading catamaran propagated to the following catamaran, they attenuated the energy of the regular waves, causing weaker wave disturbances for the following catamaran. Consequently, the pitch and heave motion responses of the following catamaran were reduced. The waves generated by the lead catamaran created complex pressure distributions around the catamaran, especially near the bow and stern. In contrast, the pressure distribution around the following catamaran was more stable, resulting in smaller pitch and heave motion responses, because most of the waves were absorbed and dispersed by the lead catamaran.

Figure 13 shows a comparison of the total resistance coefficients of a three-ship system at different longitudinal distances with that of a single ship. Figure 14a,b correspond to comparisons of heave and pitch amplitudes, respectively. The figure shows that the three-ship system exhibited a trend consistent with the two-ship system, with ship 3 showing smaller overall variations compared to ship 2. At $ST = 1 L_{pp}$, the resistance coefficient of the follower catamaran decreased significantly, with ship 2 experiencing a maximum reduction of 35.24% and ship 3 experiencing a maximum reduction of 46.44%. Both the heave

and pitch motions of the two follower catamarans decreased as the longitudinal distance increased, with ship 3 showing the greatest reduction in motion response. Similar to the two-catamaran system, the third catamaran in the three-catamaran system experienced further reduced pitch and heave motions after the waves were attenuated by the first two catamarans. Additionally, due to the shielding effect of the lead catamaran, the flow velocity around the third catamaran decreased, and the wave disturbances were reduced, causing a redistribution of surface pressure and a further reduction in resistance compared to the second catamaran.

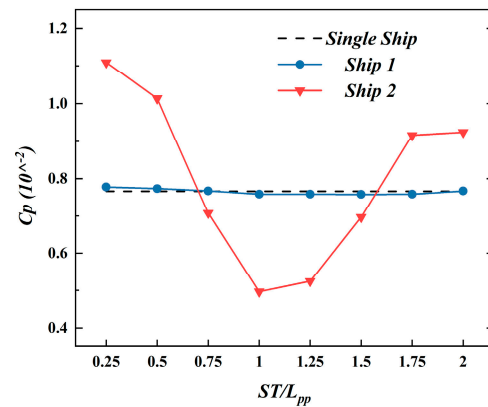


Figure 11. Comparison of the total resistance coefficients for each catamaran in the two-ship system.

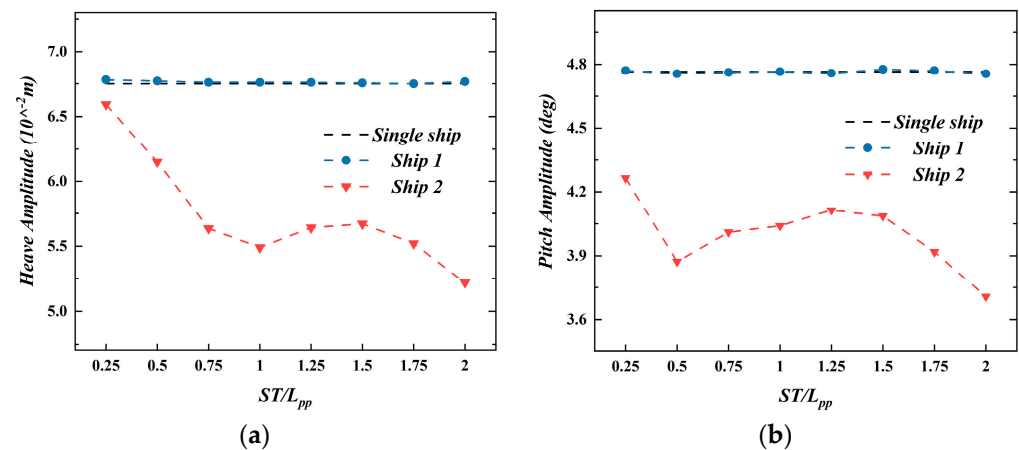


Figure 12. Comparison of the motion characteristics for each catamaran in the two-ship system. (a) Heave amplitude and (b) pitch amplitude.

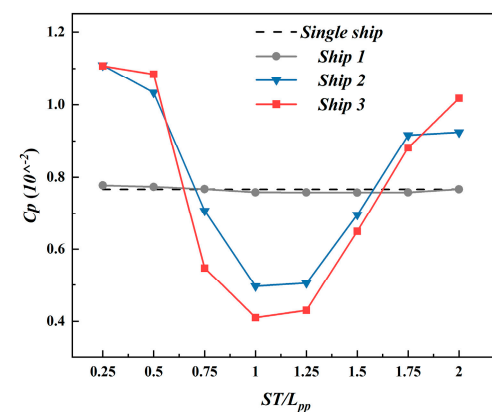


Figure 13. Comparison of the total resistance coefficients for each catamaran in the three-ship system.

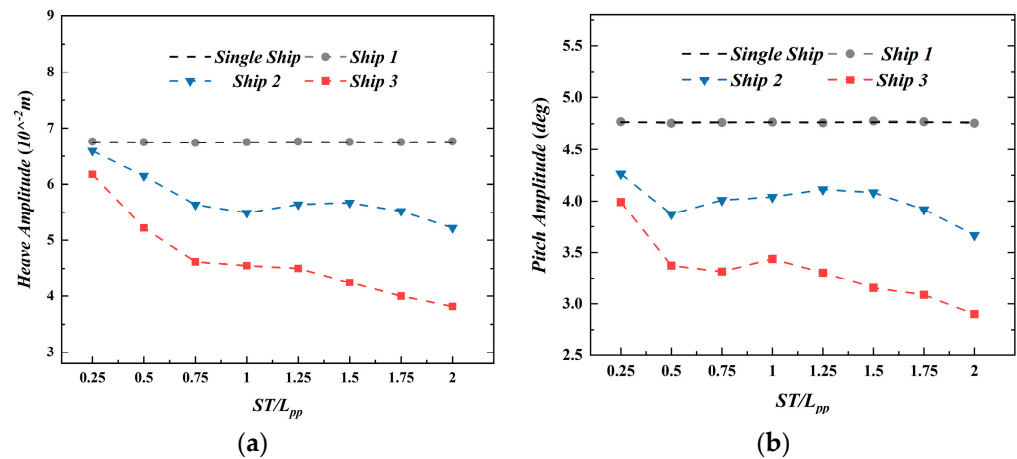


Figure 14. Comparison of the motion characteristics for each catamaran in the three-ship system. (a) Heave amplitude and (b) pitch amplitude.

The transom stern is widely used in high-speed hulls to reduce resistance [33]. The sketch of the wave profile along the centerline ($Y/L = 0$) is shown in Figure 15. The red dashed circle in the figure indicates the wave crest generated by the catamaran's wake. At high speeds, a depression formed behind the centerline of the transom stern, which then rose to form a rooster tail. The rooster tail was caused by the interference of two Delft-372 pieces, which created a more pronounced rooster tail on the centerline. It can be reasonably assumed that when the follower ship is in the rooster tail created by the leading ship, the resistance will increase significantly.

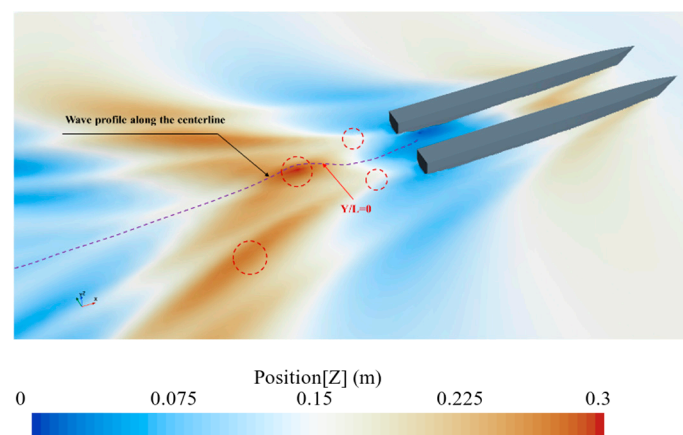


Figure 15. The sketch of the wave profile along the centerline.

Figure 16 illustrates the free surface deformations of tandem formation at different longitudinal distances. Both catamarans generated stern waves in the process of moving forward. The stern wave from the leader catamaran disturbed the stern wave from the follower catamaran, affecting the flow field around the follower. The surface pressure on the follower catamaran was redistributed, altering the wave-making resistance. This led to changes in the surrounding flow field, resulting in a redistribution of surface pressure on the follower catamaran and subsequent changes to its wave-making resistance. The resistance and motion of the leading catamaran were observed to be similar to those when sailing alone, mainly due to the high speed of the Delft-372 catamaran. In this scenario, waves from the trailing catamaran were propagated by splash effects, causing minimal disruption to the wake of the leader catamaran. Consequently, the flow field around the leader catamaran was observed to be less affected by the presence of the trailing catamaran.

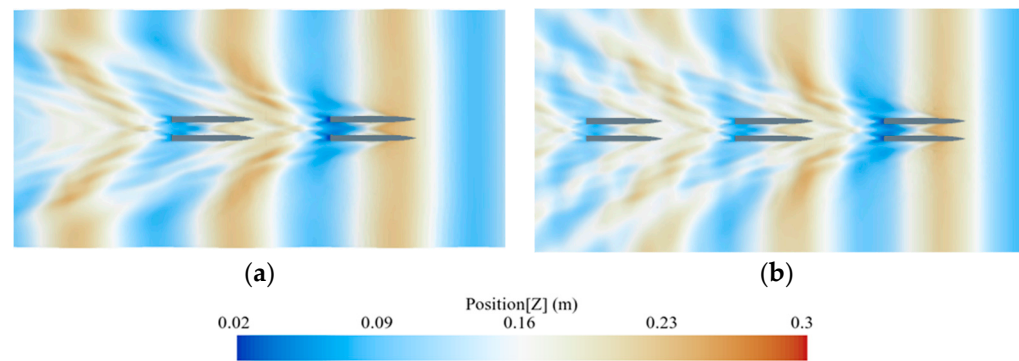


Figure 16. The free surface deformations of tandem formation in the two- and three-ship systems. (a) Two-ship system and (b) three-ship system.

In the range of 0.25 to $0.5 L_{PP}$, the follower catamaran moved in the rooster tail and the peak of the stern wave generated by the leader. The catamaran's forward speed was higher than the surrounding flow speed, resulting in additional thrust from the wave crest. This created a pressure differential across the bow and stern of the catamaran, increasing its wave-making resistance compared to both the leader catamaran and a single catamaran. Consequently, the overall resistance increased by up to 44.65%.

As the longitudinal distance increased, the follower catamaran moved closer to the trough region of the leader catamaran's wake. The leader catamaran's strong shielding effect on the incoming flow reduced the incident velocity on the follower. Simultaneously, the follower catamaran's wave resistance was significantly reduced by the suction force from the leader catamaran's wake trough, noticeably reducing its pitch and heave amplitudes. A reduction in resistance of up to 36.61% was achieved, indicating a significant difference. These observations demonstrated the significant influence of both wave crests and troughs.

4.3. Lateral Formation

The total resistance values of the follower catamaran at various lateral and longitudinal distances are shown in Figure 17, with a longitudinal distance of '0' indicating that the catamarans were in tandem formation. Overall, the total resistance of the catamarans in lateral formation tended to increase compared to tandem formation, except for $ST = 1.25 L_{PP}$. The total resistance was significantly influenced by the longitudinal distance, showing a variation pattern almost identical to that of the tandem formation. Increasing the longitudinal distance, ST , from $0.5 L_{PP}$ to $1.25 L_{PP}$ resulted in a decrease and subsequent increase in the follower catamaran's total resistance, indicating a significant drag reduction effect at $ST = 1.25 L_{PP}$, reduced by up to 44.54% at $SP = 0.75B$. When $ST = 0.25 L_{PP}$ or $0.5 L_{PP}$, the total resistance coefficient of the follower catamaran in lateral formation exceeded that of a single catamaran, placing it in the least favorable position. It was observed that irregular variations in the resistance of the follower catamaran occurred in response to changes in the parallel distance, depending on its specific position relative to the leader catamaran.

The pitch and heave values of the follower catamaran at different parallel and longitudinal distances are shown in Figure 18, with a longitudinal distance of '0' indicating a tandem formation. Generally, it was observed that the pitch and heave responses increased in the lateral formation compared to the tandem formation, except for $ST = 1 L_{PP}$. In line with the observed trend in total resistance changes, as ST increased from $0.5 L_{PP}$ to $1 L_{PP}$, the pitch and heave responses of the follower catamaran initially decreased and then increased, indicating a significant damping effect at $ST = 1 L_{PP}$.

Figure 19 illustrates the free surface deformations of lateral formation at different longitudinal and lateral distances. When the follower catamaran was positioned at the crest of the leader catamaran's wake, within a region of high lateral pressure, an asymmetrical flow was induced around the vessel. This flow created a pressure differential between the port and starboard sides, and a steep pressure gradient, generating suction

forces on the catamaran. Simultaneously, the thrust from the wake crest contributed to a significant pressure differential between the bow and stern, leading to an increase in form-drag resistance. Under these combined influences, when the follower catamaran was situated at the crest of the leader catamaran's wake, both its resistance and motion response dramatically increased, with detrimental effects. Therefore, it is recommended that the follower catamaran be positioned away from the crest of the leader catamaran's wake during navigation.

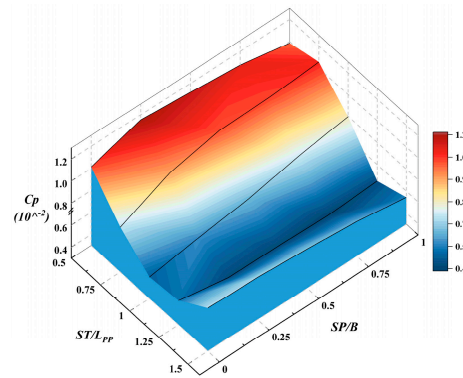


Figure 17. Comparison of the total resistance coefficients for each catamaran in lateral formation.

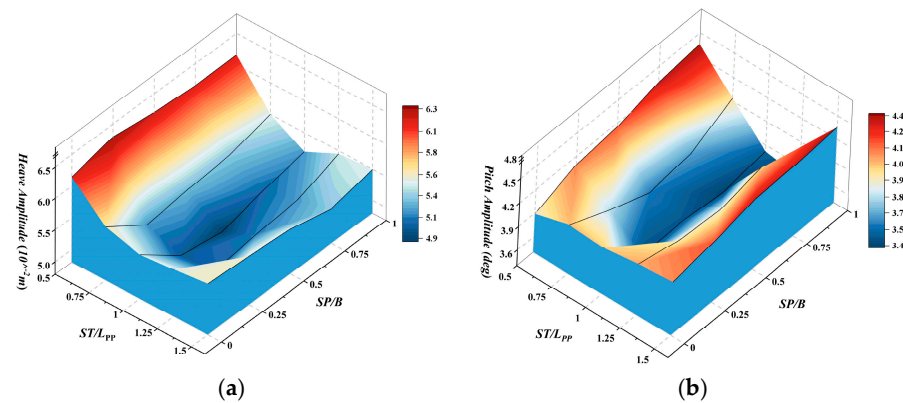


Figure 18. Comparison of the motion characteristics for each catamaran in lateral formation. (a) Heave amplitude and (b) pitch amplitude.

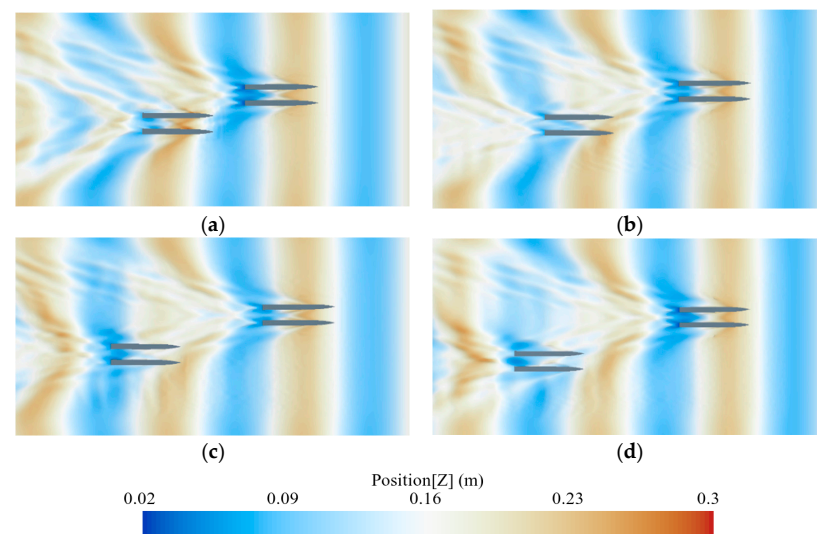


Figure 19. The free surface deformations of lateral formation in different longitudinal and lateral distances: (a) $ST/L_{pp} = 0.5$, $SP/B = 0.25$; (b) $ST/L_{pp} = 1$, $SP/B = 0.5$; (c) $ST/L_{pp} = 1.25$, $SP/B = 0.75$; (d) $ST/L_{pp} = 1.5$, $SP/B = 1$.

5. Conclusions

Research has found that multiple vessels sailing in different formation configurations can achieve drag and roll reduction effects through wave interference under specific layouts. In this work, the Delft-372 catamaran was utilized to investigate the feasibility of drag reduction and roll mitigation for catamaran formation sailing in waves, analyzing the effects of three different formation configurations and varying spacings. This study analyzed the impact of various formation configurations and different lateral and longitudinal distances on the resistance, pitch, and heave motions of catamarans, comparing these effects with those on a single vessel. The feasibility of drag reduction and roll stabilization for catamarans navigating in formation through waves was demonstrated, and strategies for leveraging advantageous positions to achieve positive outcomes were analyzed. The following conclusions were drawn from the study and analysis:

- (1) In parallel formation, the geometric symmetry resulted in nearly identical resistance and motion responses for the two catamarans. The resistance coefficient of the catamaran increased by up to 13.67% at a transverse spacing of $0.25B$. As the transverse spacing increased, the total resistance coefficient, lateral force, and pitch and heave motion amplitudes of the catamaran continuously decreased. When $SP \geq 1B$, the error in the total resistance coefficient of the catamaran, compared to a single catamaran, was within 5%. Therefore, in lateral formation, the transverse spacing between the two ships should be at least $1B$.
- (2) In tandem formation, there were no significant changes in the total resistance coefficient, pitch, and heave response of the leader catamaran. The optimal distance for resistance benefits was at $ST = 1 L_{pp}$ where the total resistance coefficient of ship 2 in the two-ship system decreased by up to 36.61%, and ship 2 and ship 3 in the three-ship system decreased by up to 35.24% and 46.44%, respectively. In conclusion, optimal efficiency was achieved at $ST = 1 L_{pp}$. When the follower catamaran moved closer to the trough region of the leader catamaran's wake, the strong shielding effect of the leader catamaran reduced the incident velocity of the fluid on the follower catamaran, thereby decreasing its pitch and heave motion amplitudes. Additionally, the suction at the trough of the wake of the leading catamaran altered the surface pressure distribution on the following vessel, significantly reducing its resistance.
- (3) In lateral formation, due to the influence of the leader catamaran's wake, changes in the follower catamaran's resistance and motion response were closely related to its position. Overall, longitudinal distance had the most significant effect, with the greatest reduction in the resistance occurring at $ST = 1.25 L_{pp}$ and the greatest reduction in pitch and heave responses at $ST = 1 L_{pp}$. The influence of lateral distance was related to the positions of the wake troughs and crests. The crest of the wake of the leader catamaran provided thrust, while the trough provided suction. This caused significant changes in the flow field around the follower catamaran, leading to asymmetric flow around it and a redistribution of surface pressure. Utilizing beneficial wave interference to fully reduce the surface pressure difference can effectively decrease wave resistance.

Through the numerical analysis in this paper, certain suggestions can be made for the arrangement and the spacing of the formations in order to achieve a reasonable sailing layout. It is necessary to maintain an appropriate distance between ships to ensure sufficient safety. From the perspective of resistance and motion response, the tandem formation was superior to parallel and lateral formations, with the best performance observed when the longitudinal distance was $1 L_{pp}$. In general, the follower catamaran should ideally be positioned in the trough of the lead catamaran's stern wave during the navigation process.

This paper only analyzed catamarans' different formation layouts and spacing at a single speed under wave conditions. Due to limitations in research costs and time, further investigation into formations with greater distances and higher speeds has not yet been explored. The current study focused on the impact of the leading ship's wake on the trailing ship under wave conditions but lacked research on formations composed of

different configurations and varying numbers of ships. The future study will consider the formation of three or more catamarans at different speeds, with the objective of exploring the appropriate formation layouts and greater distances. Future research on the formation will also focus on a catamaran with different body spacing and analyze the influence of the body spacing of the catamaran on the hydrodynamic performance of the formation.

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